

From Implicit Guarantees to Fiscal Reallocation: A City-Level Stress Test of Full LGFV Exit in China

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Abstract

China's program to separate local government financing vehicles (LGFVs) from government financing raises a forward-looking fiscal question: where would adjustment occur if implicit guarantees no longer supported rollover? This paper develops a transparent city-level stress test rather than an ex post causal evaluation. We combine a 2015–2023 panel for 337 prefecture-level cities with a public-LGFV-bond maturity schedule for 2025–2028. The exercise maps a full-exit refinancing shock into revenue composition, protected expenditure, budget buffers, and residual restructuring needs. In 2025, public LGFV bond maturities total RMB 3.71 trillion. The simulated financing gap ranges from RMB 1.04 trillion under orderly exit to RMB 2.51 trillion under strict exit. The composition of adjustment is more revealing than the aggregate gap: relative to orderly exit, strict exit lowers simulated tax revenue by 3.0 percent, government-fund revenue by 11.5 percent, and land-transfer revenue by 17.0 percent. The largest absolute gaps occur in high-debt, economically large cities, while the most fragile configuration is high land dependence combined with weak transfer buffers. City profiles show why aggregate regional averages are insufficient: Chongqing faces the largest western illustrative maturity wall but is fully absorbed by modeled buffers, whereas Huzhou and Kunming retain positive residual restructuring needs under strict exit. The results imply that LGFV exit is primarily a problem of fiscal reallocation, land-finance exposure, and differentiated restructuring capacity rather than a uniform regional collapse.

Keywords: Local government financing vehicles; implicit guarantees; local public finance; land finance; fiscal stress testing; China

1. Introduction

Local government financing vehicles (LGFVs) have long linked China's local fiscal system to infrastructure investment, land development, and bank and bond finance. Their role is difficult to separate from the underlying fiscal architecture: a city with limited own-source revenue can finance public-investment activity through an LGFV, while land, expected fiscal transfers, and the perceived support of the local state help sustain the vehicle's creditworthiness. This arrangement has expanded investment capacity, but it has also blurred the boundary between corporate debt and public contingent liabilities [1–4]. The policy direction is to contain new hidden debt, reduce the role of implicit guarantees, and move financing platforms toward market-oriented operation. In parallel, debt-replacement and refinancing tools have been used to lower near-term rollover pressure. These developments motivate a question that is fundamentally forward-looking rather than retrospective: *if LGFVs no longer perform a government-financing function, how would the resulting refinancing constraint be absorbed by local fiscal systems?* The issue is not simply whether a city has a large LGFV debt stock. It is whether a city can shift from an implicit-guarantee financing model toward a system in which debt service, project finance, transfers, fiscal buffers, and expenditure priorities are more clearly separated.

The question cannot be answered by equating LGFV debt with statutory local-government debt. LGFVs are legally distinct entities, many debts are supported by project cash flow or restructuring, and the observed fiscal response may involve debt extension, asset operations, market refinancing, transfers, or investment reprioritization. Nor can the question be settled by ranking cities using debt stocks alone. A city may have a large LGFV balance sheet but also a broad tax base, deep local finance, and assets that facilitate restructuring. Another city may have a smaller debt stock but a narrow revenue base, high dependence on land-related revenue, and few external buffers. The relevant object is therefore a *city-specific fiscal absorption problem*. This paper develops a city-level stress test for that problem. It is deliberately not a difference-in-differences study of a completed reform [5–9]. A nationwide full exit has not been observed over a sufficiently long post-policy horizon, and the exercise does not claim to identify

the causal effect of an implemented policy. Instead, we combine historical fiscal data with a forward maturity schedule for public LGFV bonds and ask how different, explicit refinancing assumptions map into fiscal resources and protected expenditures. The approach is closer to a public-finance stress test: it provides conditional, scenario-dependent estimates that make the assumed channels visible.

The paper makes four contributions. First, it harmonizes a city-year panel for 337 prefecture-level and above cities from 2015 through 2023, combining fiscal revenue and expenditure items, government-fund and land-transfer revenue, explicit government debt, LGFV debt, macroeconomic controls, and a deduplicated public-bond maturity schedule for 2025–2028. Second, it traces the simulated shock through particular fiscal items—tax revenue, government-fund revenue, land-transfer revenue, transfers, rigid expenditure, flexible expenditure, debt interest, and a residual restructuring need—rather than producing a single risk score. Third, it distinguishes debt scale from fiscal fragility using regional groups, coastal status, land and transfer dependence, and four fiscal–debt archetypes. Fourth, it brings the national exercise down to the city level through illustrative profiles in every major region. The headline result is not a prediction of generalized fiscal collapse. Public LGFV bond maturities peak at RMB 3.71 trillion in 2025 and fall thereafter. Depending on the assumed market refinancing rate, the 2025 public-bond financing gap ranges from RMB 1.04 trillion to RMB 2.51 trillion. Under the strict scenario, the stress-test hierarchy allocates most of the gap to modeled budget buffers and limited support, leaving a residual of RMB 0.058 trillion, or 2.3 percent of the aggregate gap. This outcome is conditional on the assumed buffers and cannot be read as evidence that non-bond LGFV liabilities are safe. Its substantive implication is sharper: when basic services and debt interest are protected, the first-order fiscal incidence of full LGFV exit is concentrated in government-fund and land-finance channels, not uniformly across budget items [10–16].

The contribution summary of our paper is as follows:

- We develop a transparent city-level fiscal stress-test framework for examining the hypothetical full exit of LGFVs from government financing. Rather than treating LGFV debt as an immediate fiscal liability or claiming a causal policy effect, the framework maps observed public-bond maturities into scenario-specific refinancing gaps and allocates these gaps across budget buffers, limited upper-level support, flexible expenditure adjustment, and residual restructuring needs.
- We construct a harmonized city-level fiscal–debt dataset covering 337 prefecture-level and above cities from 2015 to 2023, and combine it with a deduplicated public LGFV bond maturity schedule for 2025–2028. This integrated dataset allows us to jointly examine fiscal revenue composition, land-finance dependence, transfer capacity, debt exposure, and the observable timing of refinancing pressure.
- We show that the consequences of LGFV exit are primarily a problem of differentiated fiscal reallocation rather than uniform fiscal collapse. The simulated pressure is concentrated in government-fund and land-transfer revenue channels, while fiscal vulnerability depends not only on maturity scale but also on the joint configuration of land dependence, transfer buffers, refinancing capacity, and residual fiscal space. These results imply that transition policies should distinguish large-scale rollover needs from genuinely fragile fiscal configurations and target support toward residual restructuring pressure rather than debt scale alone.

2. Institutional setting and conceptual framework

2.1. LGFVs, implicit support, and fiscal reallocation

China’s local fiscal system combines substantial expenditure responsibilities with uneven own-source revenue. LGFVs emerged as a mechanism through which local governments could finance public-investment projects and land development outside conventional budget constraints. The underlying credit arrangement has often rested on a combination of government ownership, asset injections, land-related revenues, bank relationships, and expectations of support. The literature emphasizes both the developmental role of this system and the risks created by contingent fiscal exposure.

A full separation of LGFVs from government financing changes the rollover problem but does not mechanically convert all vehicle debt into budget expenditure. For a city i in year t , the public-bond refinancing gap under scenario s is

$$Gap_{it}^s = Maturity_{it} (1 - \rho_i^s), \quad (1)$$

where $Maturity_{it}$ is the principal of publicly observed LGFV bonds maturing in that year and ρ_i^s is the scenario-specific market refinancing rate. We intentionally restrict $Maturity_{it}$ to public bonds with observed maturity dates. Bank loans, nonstandard credit, supplier arrears, and other obligations are not treated as known annual maturities. The measured maturity wall is therefore a lower-bound cash-flow challenge, not an estimate of all contingent obligations.

Table 1: Historical fiscal–financing calibration, first differences

Outcome	$\Delta \log LGFV \text{ debt}$	$\Delta \log \text{gov. debt}$	$\Delta \log GDP$	N
General-budget revenue	-0.006	0.040	0.303	2,688
General-budget expenditure	0.023	-0.013	0.094	2,688
Government-fund revenue	0.137	-0.093	0.099	2,668
Land-transfer revenue	0.123	0.022	0.029	2,636
Flexible expenditure	0.016	-0.038	0.140	2,688

Notes: Each row is a city-level first-difference OLS regression using the 2015–2023 panel. The table is descriptive: coefficients are used to discipline relative scenario sensitivity and are not causal estimates of a completed LGFV-exit policy. The R^2 values are low in the underlying regressions, reinforcing this interpretation.

The gap is allocated through a conservative fiscal absorption hierarchy:

$$Gap_{it}^s = B_{it}^s + T_{it}^s + F_{it}^s + R_{it}^s, \quad (2)$$

where B is available budget-buffer absorption, T is limited upper-level support, F is flexible-expenditure adjustment, and R is the residual restructuring need. R is not an assumed default. It is the portion that the scenario cannot absorb through the first three channels and would therefore require restructuring, extension, asset operations, or other arrangements. The ordering is intentionally restrictive: the model protects a narrow core of education, social security and employment, health, general public services, and debt-interest expenditure before allowing flexible project expenditure to adjust.

This structure yields three empirical implications. First, the revenue side should be particularly exposed where LGFV activity, land development, and government-fund finance are tightly connected. Land-transfer revenue and government-fund revenue should respond more strongly to stress intensity than tax revenue. Second, debt maturity scale determines the size of the immediate challenge, while fiscal self-sufficiency, transfers, land dependence, and financial depth determine the city’s ability to absorb it. Third, the most informative unit of analysis is not an administrative region alone but a fiscal configuration: high debt may be manageable when buffers are large, while a smaller maturity schedule can be difficult where revenue is narrow and financing capacity is limited.

2.2. Historical calibration and the scope of inference

We use historical city-level co-movements to discipline the stress test, not to claim causal elasticities. For fiscal outcome Y_{it}^k , we estimate a dynamic reduced-form relationship of the form

$$\Delta \log Y_{it}^k = \beta_k \Delta \log LGFV Debt_{it} + \gamma_k \Delta \log Gov Debt_{it} + \delta_k \Delta \log GDP_{it} + u_{it}^k. \quad (3)$$

The estimates are descriptive. They summarize how fiscal items historically move with changes in LGFV debt, explicit government debt, and local output within cities. Their role is to indicate which fiscal margins were historically most closely related to LGFV expansion and contraction, and to discipline the relative size of scenario shocks.

Table 1 reports the core coefficients. The historical association is stronger for government-fund and land-transfer revenue than for general-budget revenue: the estimated LGFV-debt elasticities are 0.137 and 0.123 for the former two outcomes, compared with -0.006 for general-budget revenue. Flexible expenditure has a small positive association of 0.016. These regressions have low explanatory power, which is precisely why we do not extrapolate them as a structural model or as a causal policy effect. They nevertheless motivate a scenario design in which land and fund revenues receive larger shocks than tax revenue, and in which expenditure adjustment is treated as a constrained absorption channel rather than a mechanical proportional response.

3. Data, measurement, and predictive discipline

3.1. City fiscal panel and public-bond maturity schedule

The main panel covers 337 cities from 2015 through 2023, yielding 3,033 city-year observations. It merges cleaned annual fiscal accounts with debt, macroeconomic, real-estate, and financial variables. The fiscal module distinguishes general-budget revenue, tax revenue, government-fund revenue, land-transfer revenue, and a transfer-revenue proxy. It also distinguishes protected public-service items—education, social security and employment, health, and a limited general-public-service component—from more adjustable infrastructure-related expenditure on urban and rural communities, transport, agriculture and water, and energy and environment.

Table 2: Selected variables in the city-year panel, 2015–2023

Variable	Mean	Median	p25	p75
GDP	2913.0	1557.4	824.9	3057.2
General-budget revenue	146.4	32.6	14.8	91.3
General-budget expenditure	500.6	339.4	223.5	515.5
Land-transfer revenue	129.0	37.7	9.5	105.5
LGFV interest-bearing debt	957.7	227.1	77.1	701.0
Fiscal self-sufficiency	0.263	0.095	0.050	0.227

Notes: Monetary variables are RMB 100 million. Fiscal self-sufficiency is general-budget revenue divided by general-budget expenditure. The panel contains 3,033 city-year observations for 337 cities. Variable construction follows the cleaned fiscal, debt, and bond files in the replication package.

Table 2 summarizes the main variables. The mean city has RMB 2,913 billion of GDP, RMB 146 billion of general-budget revenue, RMB 500 billion of general-budget expenditure, and RMB 958 billion of LGFV interest-bearing debt. The median is substantially lower than the mean for all scale variables, emphasizing why national aggregates conceal city heterogeneity. We use public-bond data to construct yearly issuance, maturity, coupon, and maturity concentration. Public bond maturity therefore measures the observable rollover challenge rather than total LGFV liabilities.

The data construction preserves two distinctions central for interpretation. First, LGFV interest-bearing debt and statutory local-government debt are not added together mechanically. The former is treated as a contingent fiscal exposure and the latter as an explicit fiscal constraint. Second, public-bond maturities are used to construct annual rollover needs, while total LGFV debt is used for cross-city exposure classification. Some fiscal items are harmonized proxies: non-tax revenue is constructed as general-budget revenue less tax revenue when a directly reported item is unavailable, and the transfer proxy is the positive difference between general-budget expenditure and revenue. Results should therefore be interpreted as a harmonized fiscal stress map, not an audited reconstruction of every local budget.

3.2. Fiscal capacity, city types, and a residual fiscal-space proxy

We classify cities along two transparent dimensions. LGFV exposure is total LGFV interest-bearing debt relative to GDP. Fiscal capacity increases with fiscal self-sufficiency and decreases with fiscal-gap and transfer dependence. Splitting both dimensions at their pre-stress medians creates four archetypes: strong fiscal/low debt (A), strong fiscal/high debt (B), weak fiscal/low debt (C), and weak fiscal/high debt (D). We additionally classify cities by official four-region group, coastal status, land dependence, transfer dependence, population growth, and maturity concentration.

We also construct a deliberately conservative proxy for non-core discretionary fiscal space:

$$Capacity_{it}^s = \max\{0, Resources_{it}^s - CoreFloor_{it}^s\}. \quad (4)$$

Resources are restricted to own general-budget revenue and non-earmarked transfer resources where observable. The core floor includes protected public-service spending and debt-interest expenditure. This measure does *not* identify salaries, bonuses, allowances, or any specific employee benefit. It is a residual-budget proxy: the potential fiscal room remaining after a narrow set of protected obligations. Actual compensation and benefits depend on economic classifications, earmarking, staffing, departmental obligations, and legal rules that are not observed here.

The distinction matters for the final part of the analysis. A city with positive modeled residual capacity should not be interpreted as having a verified capacity to provide discretionary personnel compensation. The indicator only means that the modeled LGFV shock does not exhaust the narrow fiscal-resource measure after the selected protected floor. Conversely, a zero indicator does not establish that basic payroll or benefits cannot be met. It simply identifies a configuration in which the stress test exhausts the specified budget margin.

3.3. Model validation

Because the exercise is predictive rather than causal, predictive discipline is important. We compare rolling forecasts using a simple dynamic panel benchmark and machine-learning alternatives during model development. The paper retains the transparent dynamic specification as the primary framework and uses out-of-sample error as a limitation check, not as a basis for selecting an opaque black box. Table 3 reports one-year-ahead rolling MAPE for 2023. GDP and general-budget expenditure are relatively predictable, at 4.2 percent and 6.4 percent, respectively. General-budget revenue, government-fund revenue, and land-transfer revenue are much less predictable, with MAPE between 61.7 and 99.4 percent.

Table 3: Rolling out-of-sample forecast performance for 2023

Forecasted item	2023 rolling MAPE	Cities
GDP	4.2%	337
General-budget revenue	61.7%	337
General-budget expenditure	6.4%	337
Government-fund revenue	63.8%	336
Land-transfer revenue	99.4%	334

Notes: MAPE is based on rolling forecasts for the final observed year. The table documents forecast difficulty, especially for land-linked revenue, and motivates reporting scenarios rather than point forecasts as policy predictions.

Table 4: 2025 aggregate results across full-exit scenarios

Scenario	Gap	Residual	Cliff cities	Δ Tax	Δ Fund	Δ Land
Orderly exit	1.04	0.017	0	0.0%	0.0%	0.0%
Benchmark exit	1.77	0.029	0	-1.0%	-4.2%	-6.4%
Support-calibrated exit	1.91	0.028	0	-1.5%	-6.2%	-9.6%
Strict exit	2.51	0.058	2	-3.0%	-11.5%	-17.0%

Notes: Gap and residual are RMB trillion. Revenue changes are relative to the orderly-exit scenario. “Cliff cities” are cities for which the simulated financing gap exceeds modeled flexible expenditure plus available budget-buffer capacity. This is a stress-test threshold, not a forecast of default.

This pattern is substantively informative. The most policy-relevant revenue margins are also the most volatile and difficult to forecast. Accordingly, the paper does not present the land-revenue projection as a precise point forecast. Instead, it uses scenario bands and emphasizes the ordering across channels: land and government-fund revenue are modeled as more exposed than tax revenue, while basic public-service expenditure is protected by construction. The city results should therefore be interpreted as relative fiscal pressure under transparent assumptions, not as an annual budget forecast.

4. Stress scenarios and fiscal accounting

We model four full-exit scenarios. The orderly-exit scenario uses a 75 percent baseline market refinancing rate; the benchmark exit uses 55 percent; the strict exit uses 35 percent; and the support-calibrated scenario begins at 50 percent and varies with fiscal capacity and debt exposure. Revenue-side shocks are deliberately larger for land and government-fund revenue than for tax revenue. Across orderly, benchmark, and strict exits, tax-revenue shocks are -1 , -2 , and -4 percent; government-fund shocks are -4 , -8 , and -15 percent; and land-transfer shocks are -6 , -12 , and -22 percent, respectively. Non-tax revenue is allowed a small positive offset, reflecting the possibility of administrative and non-tax adjustment; this is a stress-accounting assumption, not a welfare claim.

The scenario matrix does not assume that every city has the same refinancing probability. A city’s scenario refinancing rate is anchored to its historical issuance-to-maturity experience, then adjusted by fiscal capacity, LGFV exposure, transfer dependence, and observed financial depth. The design does not estimate a true market probability of refinancing. It operationalizes a comparative question: how much more of the observable maturity wall must be absorbed elsewhere when market rollover is progressively less available?

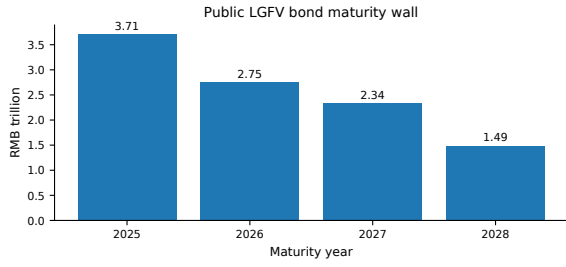
Table 4 reports 2025 aggregate implications. Because scenarios differ in refinancing and fiscal-resource assumptions, the table should be read horizontally as a sensitivity analysis. The orderly-exit scenario provides the least severe full-exit benchmark; the strict exit is intentionally adverse. The support-calibrated scenario tests whether limited differentiated support alters the composition of adjustment.

5. Aggregate results

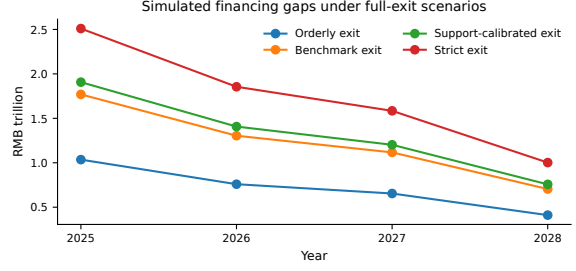
5.1. The maturity wall and the aggregate financing gap

Figure 1 shows a pronounced but declining public-bond maturity wall. Public LGFV bond maturities total RMB 3.71 trillion in 2025, then fall to RMB 2.75 trillion in 2026, RMB 2.34 trillion in 2027, and RMB 1.49 trillion in 2028. This declining profile does not imply that LGFV risk disappears. It simply indicates that the publicly observed bond component creates its largest near-term cash-flow challenge in 2025. The profile also reinforces a practical point: a transition policy that is administratively gradual can still confront a sharp near-term refinancing concentration.

The simulated financing gap mirrors the maturity wall. Figure 1 shows that the 2025 gap ranges from



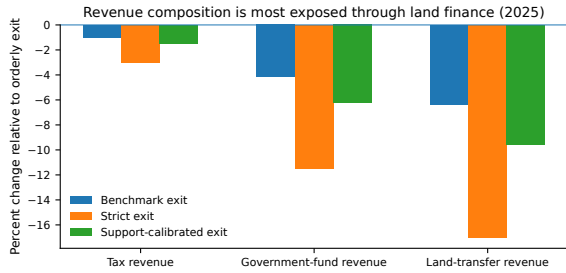
(a) Public LGFV bond maturity wall.



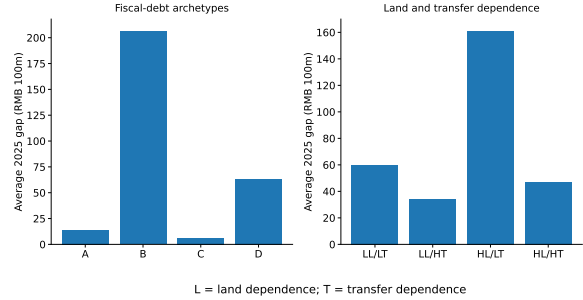
(b) Financing-gap sensitivity.

Figure 1: Public LGFV bond maturities and simulated financing gaps, 2025–2028

Notes: Amounts are aggregated across the 337 cities and reported in RMB trillion. The gap depends on each scenario's market refinancing rate and is based on publicly observed bond maturities only.



(a) Revenue composition under more severe exit assumptions.



(b) Heterogeneity in the strict 2025 scenario.

Figure 2: The fiscal incidence of full LGFV exit

Notes: Panel (a) reports percentage changes relative to orderly exit. Panel (b) reports mean city-level financing gaps in RMB 100 million. In the land/transfer labels, L denotes land dependence and T denotes transfer dependence.

RMB 1.04 trillion in orderly exit to RMB 2.51 trillion in strict exit. The benchmark exit produces a RMB 1.77 trillion gap. By 2028, the corresponding range narrows to RMB 0.41–1.00 trillion. This result is not a prediction of realized losses. It is a transparent answer to a narrower question: conditional on observed public maturities and scenario-specific refinancing rates, how much refinancing must be replaced by other fiscal or restructuring channels?

5.2. The revenue incidence is concentrated in land finance

The first substantive finding is compositional. Under strict exit, tax revenue in 2025 is 3.0 percent below the orderly-exit benchmark, government-fund revenue is 11.5 percent lower, and land-transfer revenue is 17.0 percent lower. This ranking is consistent with the historical calibration: land and fund revenues are more closely tied to the investment and development system through which LGFVs have historically operated. General-budget revenue is less directly exposed in the scenario design because it reflects a broader tax base and because the model does not assume that all local economic activity falls proportionately with project finance.

This distinction matters for fiscal adjustment. A city can maintain an apparently stable general-budget revenue path yet face a substantially weaker government-fund position. The latter has consequences for urban construction, housing, land preparation, and other project-linked activities even where basic current expenditure is protected. Thus, a stress test centered only on general-budget income would understate the importance of land finance; a stress test centered only on debt stocks would miss the revenue channel through which investment-oriented fiscal adjustment is likely to occur.

5.3. Regional patterns: scale is not fragility

Table 5 reports strict-exit results for 2025. Eastern cities have the largest average financing gap in level terms, followed by central, western, and northeastern cities. Coastal cities similarly account for a disproportionate share of aggregate maturities because they include large issuers and cities with large debt stocks. This pattern is a scale result, not a judgment that eastern or coastal cities are fiscally weaker. Their income bases, financial depth, and modeled buffers can also be larger.

The four-archetype classification makes the point more precisely. Type B cities—strong fiscal capacity

Table 5: Heterogeneity under the strict full-exit scenario, 2025

Group	Cities	Mean gap	Mean gap/GDP
<i>Panel A. Region</i>			
Northeast	36	0.10	0.55%
East	88	1.51	1.85%
Central	82	0.71	1.95%
West	131	0.43	1.83%
<i>Panel B. Fiscal-debt archetype</i>			
A: strong fiscal, low debt	81	0.15	0.36%
B: strong fiscal, high debt	88	2.06	2.65%
C: weak fiscal, low debt	87	0.07	0.38%
D: weak fiscal, high debt	81	0.63	2.49%

Notes: Mean gap is reported in RMB billion. Mean gap/GDP is the city-average ratio. Fiscal-debt archetypes are based on pre-stress fiscal capacity and LGFV exposure. East, central, west, and northeast follow the standard four-region classification.

Table 6: Illustrative city profiles under the strict full-exit scenario, 2025

Region	City	Type	Maturity (RMB bn)	Gap (RMB bn)	Gap/GDP (%)	Refi. (%)	Land ratio	Residual (RMB bn)
East	Tianjin	B	123.3	90.4	5.1%	26.7%	0.90	0.0
	Huzhou	B	73.2	67.4	15.2%	8.0%	0.90	34.1
Central	Changsha	B	64.0	41.3	2.6%	35.5%	1.20	0.0
	Taiyuan	A	20.9	12.4	2.0%	40.6%	0.85	0.0
West	Chongqing	B	211.4	151.7	4.6%	28.2%	1.01	0.0
	Kunming	B	78.9	52.7	6.2%	33.2%	1.66	13.3
Northeast	Changchun	B	26.4	16.2	2.1%	38.5%	0.95	0.0
	Harbin	A	9.3	5.9	1.0%	36.9%	0.57	0.0

Notes: Type A is strong fiscal capacity/low LGFV exposure; type B is strong fiscal capacity/high LGFV exposure. The land ratio is the model’s land-dependence measure, not a share of freely disposable fiscal resources. The cases are chosen to contrast scale and absorption mechanisms within each region; they are not a representative sample or a ranking of actual fiscal conditions.

but high LGFV exposure—face the largest average strict-exit gap, while type D cities—weak fiscal capacity and high LGFV exposure—have smaller average absolute gaps but potentially weaker absorption capacity. The group that combines high land dependence with low transfer dependence is particularly exposed: its average 2025 strict-exit financing gap is RMB 161.2 billion, with a mean residual gap of RMB 10.7 billion and a 2.5 percent fiscal-cliff share. The interpretation is not that transfer dependence is intrinsically good or that land finance is intrinsically bad. Rather, a city that relies heavily on land-related resources but cannot replace a shock through stable transfers has fewer budgetary margins when property-linked revenue weakens.

6. City profiles within regions

Regional averages provide a useful first map but can hide important within-region variation. This section therefore examines two illustrative cities in each major region. The cases are selected to contrast scale and absorption within a region; they are not statistically representative city case studies, and their modeled quantities are not observations of realized distress. Table 6 reports the core 2025 strict-scenario quantities. The figures should be interpreted in the stress-test accounting units: observed public-bond maturities determine the shock, the scenario refinancing rate determines the initial gap, and the absorption hierarchy allocates the gap among buffers, limited support, flexible adjustment, and a residual restructuring need.

6.1. Eastern cities: large maturity scale versus concentrated fragility

The eastern cases illustrate why a regional label is insufficient. Tianjin has RMB 123.3 billion of observed 2025 public LGFV bond maturities and a strict-scenario gap of RMB 90.4 billion, equivalent to 5.1 percent of modeled GDP. The assumed market refinancing rate is 26.7 percent. Yet the stress-test hierarchy absorbs the entire gap through the modeled buffer component, leaving no residual restructuring need. This result does not establish that Tianjin can freely deploy the calculated buffer; it shows only that, under the paper’s narrow resource definition and protection floor, maturity scale alone does not produce a residual gap.

Huzhou provides the contrasting case. Its observed maturity is lower at RMB 73.2 billion, but its strict-scenario refinancing rate is only 8.0 percent and the resulting RMB 67.4 billion gap equals 15.2

percent of modeled GDP. The model assigns RMB 26.4 billion to budget buffers, RMB 6.3 billion to limited support, and only RMB 0.6 billion to flexible expenditure, leaving RMB 34.1 billion as a residual restructuring need. Huzhou is one of the two cities that crosses the paper’s fiscal-cliff threshold in 2025. The contrast with Tianjin illustrates the paper’s central point: a smaller maturity wall can be more difficult to absorb when refinancing capacity and available buffers are limited.

For context, the broad east also includes cities with even larger absolute maturity schedules. Nanjing’s strict-scenario gap is RMB 153.4 billion, the largest illustrative eastern value in the full city output. However, the fiscal-buffer fields required for the decomposition are incomplete for that observation, so we do not use it as an absorption case. Reporting this limitation is important: the maturity schedule is most complete for public bonds, while some fiscal components remain unevenly observed or imputed.

6.2. Central cities: provincial-capital scale and a lower-debt comparison

The central cases are Changsha and Taiyuan. Changsha has RMB 64.0 billion in observed maturities and a RMB 41.3 billion strict-scenario gap, equal to 2.6 percent of GDP. It is classified as type B, with high LGFV exposure but stronger fiscal capacity in the pre-stress typology. Under the model, its buffer capacity absorbs the entire measured gap. In contrast, Taiyuan is classified as type A, with lower LGFV exposure. Its RMB 20.9 billion maturity schedule yields a RMB 12.4 billion gap, or 2.0 percent of GDP, which is also fully absorbed by the modeled buffer.

The comparison does not imply that Changsha and Taiyuan face identical policy problems. Changsha’s land-dependence ratio is higher, at 1.20 versus 0.85, and its maturity scale is more than three times as large. The result instead shows a basic accounting principle: a higher maturity burden does not necessarily translate into a residual gap when the scenario assigns sufficient buffer capacity. In a more restrictive liquidity or earmarking assumption, Changsha’s result would become more sensitive than Taiyuan’s. The city pair therefore underscores why the paper reports multiple scenarios rather than a single deterministic forecast.

6.3. Western cities: aggregate scale and land-linked residual pressure

The western cases distinguish scale from fiscal incidence. Chongqing has the largest illustrative 2025 maturity schedule in the sample table, RMB 211.4 billion, and a strict-scenario financing gap of RMB 151.7 billion, equal to 4.6 percent of GDP. The implied refinancing rate is 28.2 percent. In the paper’s accounting, the estimated budget buffer absorbs the full measured gap. The result should not be interpreted as an assertion that a single pool of liquid fiscal resources exists; it indicates that a large absolute maturity burden can be accommodated in the model when the city is assigned a large fiscal-resource base.

Kunming provides a different pattern. Its maturity schedule is RMB 78.9 billion and its strict gap is RMB 52.7 billion, or 6.2 percent of GDP. Its land-dependence ratio, 1.66, is the highest among the eight illustrative cases. The model applies RMB 30.9 billion of budget-buffer absorption, RMB 7.0 billion of limited support, and RMB 1.5 billion of flexible-expenditure adjustment, leaving RMB 13.3 billion as a residual restructuring need. This profile is consistent with the paper’s aggregate land-finance result: the risk is not merely the initial gap, but the limited replacement of land-linked fiscal resources once a project-finance system is disrupted.

6.4. Northeastern cities: smaller maturity walls and heterogeneous fiscal bases

Northeastern cities have the smallest average maturity gaps in the regional table, but the within-region contrast remains informative. Changchun has RMB 26.4 billion of maturities and a RMB 16.2 billion strict-scenario gap, equal to 2.1 percent of GDP. It is classified as type B because its LGFV exposure is above the median despite relatively strong fiscal capacity under the constructed index. Harbin, classified as type A, has RMB 9.3 billion of maturities and a RMB 5.9 billion gap, or 1.0 percent of GDP. Both city gaps are absorbed by the modeled buffer channel.

The result should not be read as evidence that northeastern fiscal adjustment is trivial. The regional average gap is smaller partly because public-bond maturity schedules are smaller. The model does not observe all bank loans, nonstandard financing, or project arrears, and it does not claim that a city with a zero residual has no economic or demographic constraint. Rather, the cases show that the specific public-bond rollover channel does not mechanically place every northeastern city above the paper’s threshold. This is an important corrective to a simple geographic narrative.

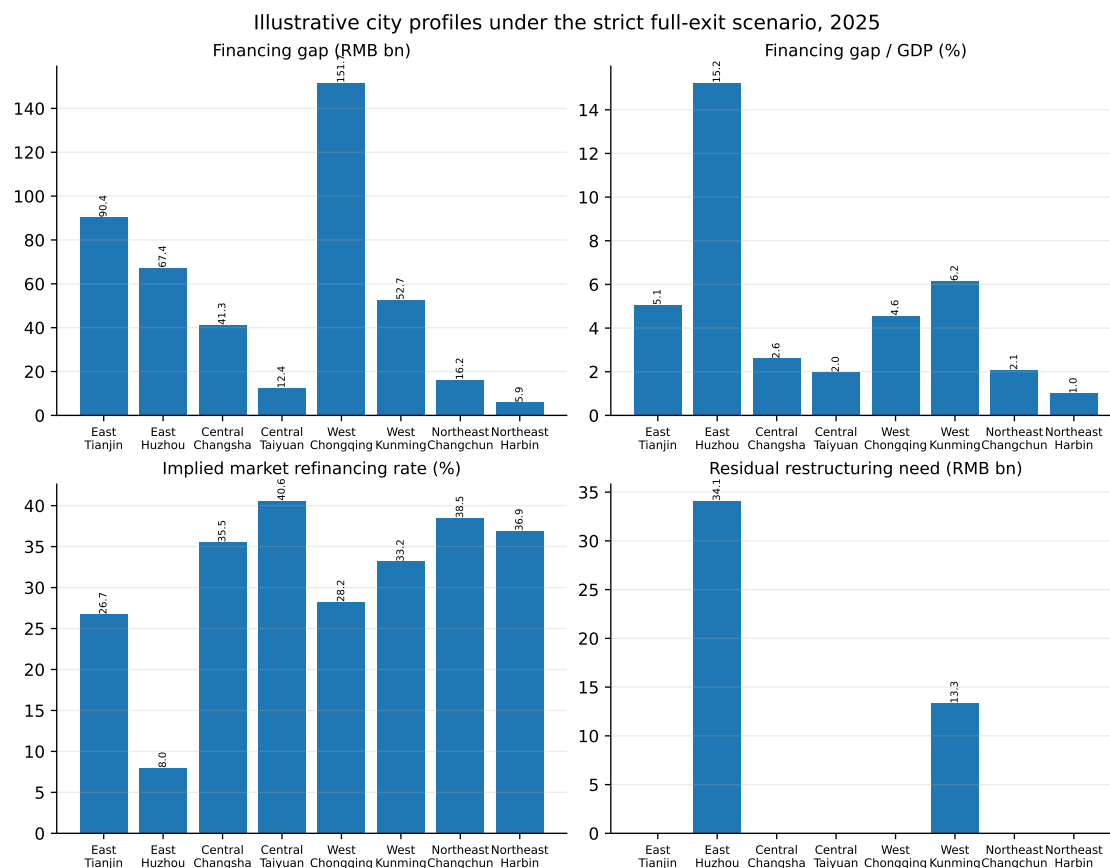


Figure 3: Contrasting city profiles under the strict full-exit scenario, 2025

Notes: The eight cities are illustrative cases selected to contrast maturity scale and absorption mechanisms within the four standard macro-regions. Amounts are RMB billion. The residual is a modeled restructuring need after the stated absorption hierarchy; it is not a forecast of default.

7. Additional findings: adjustment channels and residual fiscal space

7.1. Land dependence and debt-service crowding

Figure 4 extends the regional analysis in two directions. The left panel groups cities by high or low land dependence and transfer dependence. Under strict exit, the high-land/low-transfer group has an average financing gap of RMB 161.2 billion and an average residual of RMB 10.7 billion. By comparison, the high-land/high-transfer group has a smaller average gap of RMB 46.9 billion and no average residual in the reported 2025 scenario. The difference illustrates that land reliance alone is not sufficient to create a residual problem. What matters is the joint configuration of revenue exposure and the availability of alternative fiscal buffers.

The right panel relates debt-interest expenditure to the share of expenditure classified as flexible. It should not be read as a causal crowding-out regression. It is an accounting diagnostic: cities with high debt-service proxies and low flexible-expenditure shares have less room to absorb a refinancing shock through project reprioritization. The illustrative cases reinforce the point. Chongqing's flexible-expenditure share is comparatively large in levels, but its maturity wall is also large; Kunming combines a meaningful maturity burden with a smaller flexible-expenditure margin and therefore retains a residual after the stress hierarchy is applied. The result suggests that an exit policy should distinguish a city that needs an orderly market rollover from a city that needs restructuring support because its project-expenditure margin is limited.

7.2. A maturity wall need not become a general budget cliff

The second additional finding concerns the absorption margin. In the strict 2025 scenario, the aggregate financing gap is RMB 2.51 trillion, but the simulated residual restructuring need is RMB 0.058 trillion, or 2.3 percent of the gap. Two of 337 cities cross the fiscal-cliff threshold. The same calculation produces no cliff cities in the orderly and benchmark scenarios. Figure 5 shows why: under the model's hierarchy, a substantial share of the gap is assigned to existing budget-buffer capacity before flexible-expenditure compression or residual restructuring is invoked.

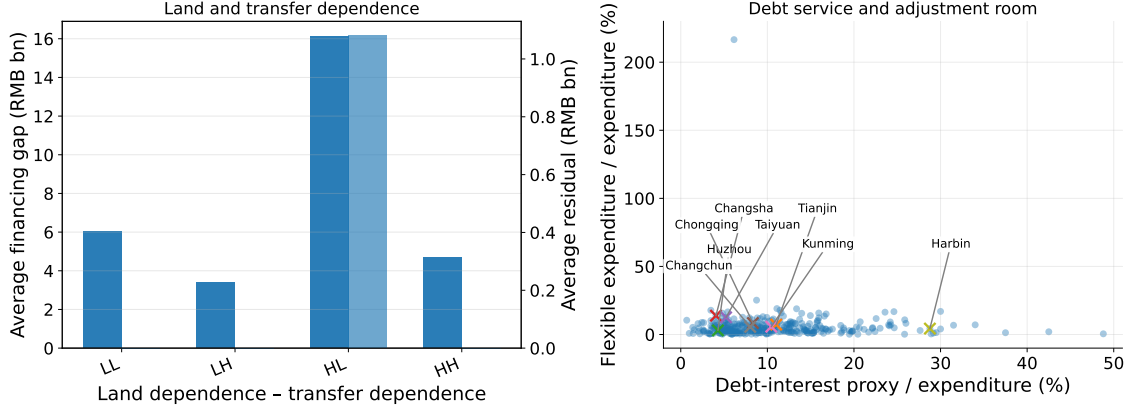


Figure 4: Land/transfer dependence and adjustment capacity under strict exit

Notes: Left panel: city-average financing gaps and residual gaps by land- and transfer-dependence groups in 2025. Right panel: a descriptive cross-section of the debt-interest proxy and flexible expenditure share. Marked cities are the illustrative city cases. Neither panel is a causal estimate.

Table 7: Illustrative city-level financing-gap absorption, strict scenario, 2025

Region	City	Gap	Buffer	Support	Flexible	Residual
RMB bn, strict 2025 scenario						
East	Tianjin	90.4	90.4	0.0	0.0	0.0
	Huzhou	67.4	26.4	6.3	0.6	34.1
Central	Changsha	41.3	41.3	0.0	0.0	0.0
	Taiyuan	12.4	12.4	0.0	0.0	0.0
West	Chongqing	151.7	151.7	0.0	0.0	0.0
	Kunming	52.7	30.9	7.0	1.5	13.3
Northeast	Changchun	16.2	16.2	0.0	0.0	0.0
	Harbin	5.9	5.9	0.0	0.0	0.0

Notes: Buffer, support, flexible, and residual are the components of equation (2). The decomposition is scenario accounting, not a realized financing plan. A zero residual does not imply the absence of liquidity, earmarking, or non-bond debt risk.

This is a conditional finding, not reassurance by assertion. The buffer is a modeled capacity derived from fiscal resources after a narrow protected expenditure floor; it is not cash on hand, and it may be unavailable because of earmarking, liquidity, project commitments, or legal restrictions. It is useful precisely because it identifies the critical assumption. If fiscal buffers cannot be mobilized at the calibrated rate, or if non-bond liabilities mature faster than public bonds imply, residual gaps rise materially. A ten-percentage-point reduction in assumed refinancing capacity raises the aggregate benchmark-exit gap from RMB 3.87 trillion to RMB 5.92 trillion in the sensitivity calculations.

Table 7 shows the same mechanism at the city level. For Tianjin, Changsha, Taiyuan, Chongqing, Changchun, and Harbin, the strict-scenario gap is fully allocated to the modeled buffer channel. For Huzhou and Kunming, buffer and support components leave positive residuals. These rows are not evidence that one city has more “cash” than another. They are a transparent accounting decomposition under common rules, which helps distinguish the scale of the initial financing gap from the constrained capacity to absorb it.

7.3. Residual fiscal space is not a personnel-compensation measure

The residual-budget analysis yields a related caution. The median strict-scenario residual capacity is RMB 35.8 billion in 2025, and 0.3 percent of cities have zero measured capacity. This should *not* be interpreted as evidence about salary increases, bonuses, allowances, or any specific category of public-sector treatment. It only says that, under the narrow accounting definition in equation (4), the simulation does not exhaust all non-core budget resources for most cities. That margin can be illusory when funds are earmarked, when the fiscal calendar creates liquidity mismatches, or when unobserved obligations are large.

The appropriate interpretation is therefore comparative. Cities with a sharply declining residual-capacity proxy under strict exit have less modeled room for non-core budget reallocation than otherwise similar cities. The result is useful for identifying where an LGFV transition may force difficult project or budget choices. It cannot identify the level or distribution of public-sector compensation, and it should not

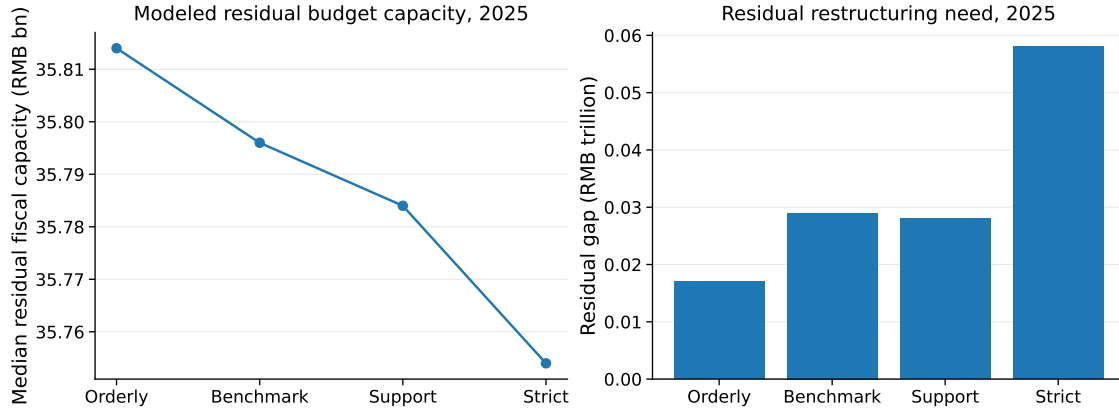


Figure 5: Modeled residual fiscal capacity and restructuring need across scenarios, 2025

Notes: The left panel reports the median city-level residual fiscal-capacity proxy after the protected expenditure floor. The right panel reports aggregate residual restructuring need. The capacity proxy is not a measure of actual compensation, bonuses, benefits, or cash balances.

be used to infer whether any city or unit can or should provide extra treatment beyond basic obligations.

8. Sensitivity, limitations, and policy interpretation

8.1. Sensitivity

The central patterns survive several transparent sensitivity checks. Restricting attention to the top 25, 33, or 50 percent of cities by debt exposure concentrates the gap but does not overturn the ranking of scenarios. Under strict exit, the top quartile of exposure accounts for a simulated RMB 5.07 trillion aggregate financing gap across the forecast years, while the top half accounts for RMB 6.53 trillion. Varying the assumed refinancing rate by ten percentage points changes the aggregate gap substantially, confirming that rollover capacity is the main uncertainty rather than a secondary detail.

The differentiated-support scenario is also informative. It reduces the 2025 residual gap modestly relative to benchmark exit, from RMB 0.029 trillion to RMB 0.028 trillion, while the total gap remains higher because the support calibration combines heterogeneous refinancing assumptions with limited support. This is not a policy recommendation to expand transfers indiscriminately. It demonstrates that support rules matter only after maturity profiles, fiscal capacity, and use restrictions are specified. In particular, a debt-proportional support allocation can direct resources toward cities with large nominal maturities while overlooking a smaller city with a higher gap relative to GDP or a positive residual after flexible expenditure is exhausted.

8.2. Limitations

Three limitations are important. First, the exercise observes public LGFV bond maturities, not the full universe of loans, trust products, supplier arrears, or contingent liabilities. The stress test is therefore conservative with respect to measured maturities but may omit other sources of pressure. Second, several fiscal items are harmonized proxies, and coverage varies across fiscal classifications. We retain the distinction between observed and model-ready panels, but the results should be viewed as an analytical dataset rather than an official fiscal audit. Third, the historical regression relationships have low explanatory power and are not causal. The paper's contribution is transparency about a plausible fiscal adjustment mechanism, not a claim that a particular scenario will materialize exactly.

A fourth limitation is especially relevant for the city profiles. The cases are generated from the same city-level dataset and scenario rules as the national results. They are intended to make the sources of heterogeneity concrete, not to offer a comprehensive fiscal diagnosis of Tianjin, Huzhou, Changsha, Taiyuan, Chongqing, Kunming, Changchun, or Harbin. Cities may have policy commitments, assets, contingent support, or financing channels that are not fully captured by public-bond data and functional fiscal accounts.

8.3. Policy interpretation

The results suggest three principles for designing LGFV exit. First, policy should distinguish maturity scale from fiscal fragility. Large coastal and eastern cities require orderly rollover and market transition

because they account for large absolute maturities. Cities with high land dependence and weak transfer buffers require attention because their fiscal incidence is less diversified. Second, fiscal support should be conditional and targeted toward residual restructuring needs rather than mechanically proportional to debt stocks. Third, protecting basic services while identifying genuinely adjustable project expenditure is preferable to treating all local spending as equally compressible.

A fourth principle follows from the city cases. The same regional policy instrument will not fit every city. A Chongqing-type scale problem calls for a credible market-transition path and monitoring of a large maturity wall. A Huzhou- or Kunming-type residual problem calls for attention to the interaction between land revenue, refinancing capacity, and fiscal buffers. A Changchun- or Harbin-type smaller observed maturity problem still requires monitoring, but the public-bond channel alone does not generate the same modeled pressure. The policy question is therefore not whether every city can be made identical after exit, but whether adjustment is allocated to channels that preserve essential public services and prevent a localized maturity shock from becoming a broader fiscal contraction.

9. Conclusion

A full exit of LGFVs from government financing would not simply shift a corporate debt stock onto public budgets. It would force a reallocation among refinancing, fiscal buffers, transfers, land-related revenues, project expenditure, and restructuring. Using a 337-city panel and a public-bond maturity schedule, this paper shows that the resulting pressure is highly uneven. The largest absolute gaps occur where debt scale and maturities are large; the most fragile fiscal configuration is high land dependence combined with weak transfer buffers. The simulated revenue adjustment falls primarily on government-fund and land-transfer channels, while the model’s protection of core services limits direct compression of basic expenditure. The regional and city profiles sharpen that conclusion. Eastern and coastal cities dominate the maturity wall in aggregate, but high nominal exposure is not equivalent to weak fiscal absorption. The illustrative city cases show that strict-scenario gaps can be fully absorbed in the model where buffers are large, whereas smaller maturity schedules can leave residual restructuring needs where refinancing capacity and fiscal margins are constrained. The paper’s main claim is deliberately narrow: these are scenario-dependent stress-test results, not realized policy effects or forecasts of statutory default. But the exercise clarifies the fiscal arithmetic that a full LGFV exit must confront. A credible transition requires more than reducing implicit guarantees; it requires a differentiated mechanism for managing maturity walls, land-finance dependence, and the fiscal capacity to absorb adjustment without eroding essential public services.

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A. Data construction, matching, and city-profile protocol

This appendix records the main construction choices behind the stress-test dataset. The fiscal panel is assembled at the prefecture-level city-year level. Annual fiscal workbooks are harmonized into common general-budget, government-fund, land-transfer, transfer-revenue, and functional-expenditure fields. Variables are retained as observed whenever possible; the model-ready panel records when a compatible value is constructed from a reported aggregate or an accounting identity. The principal examples are non-tax revenue, which is computed as general-budget revenue less tax revenue when the directly reported non-tax item is unavailable, and the transfer proxy, which is the positive difference between general-budget expenditure and general-budget revenue. These rules are transparent but should not be confused with a full official fiscal reconciliation. The public-bond module is constructed separately. Bond records are deduplicated using bond code, issuer, issuance date, maturity date, and issue size. Issuers located at county or county-level-city addresses are mapped to their prefecture-level city using an explicit administrative concordance. Provincial-level issuers are flagged rather than mechanically allocated to a city. For each city and year, the public-bond schedule aggregates issuance, maturity principal, coupon, term, and the concentration of maturities during 2025–2028. The maturity variable used in the stress test therefore contains only bonds with an observable maturity date. It is intentionally narrower than total interest-bearing LGFV debt, which is used only to classify exposure. The eight city profiles in Section 6 are selected after the national results are obtained, using a within-region contrast protocol rather than a representativeness claim. In each standard macro-region, one city is chosen for a relatively large observed maturity wall or financing gap, and one for a contrasting absorption or residual-risk profile. This yields Tianjin and Huzhou in the east, Changsha and Taiyuan in the center, Chongqing and Kunming in the west, and Changchun and Harbin in the northeast. A city is not used as an absorption illustration when the necessary fiscal-buffer components are incomplete, even when its maturity gap is large. The profiles are thus explanatory examples of the model’s accounting logic and should not be interpreted as municipal fiscal audits.

All monetary quantities in the city-profile tables are converted from the cleaned panel’s RMB 100 million unit into RMB billion. Ratios are calculated from the same city-year scenario record. The replication package retains the city-level forecast file, the cleaned fiscal–debt panel, the bond master, the city matching logs, and the table and figure data underlying the results reported here. The scenario procedure is implemented in four reproducible steps. First, the 2025–2028 public-bond maturity schedule is taken as the observed refinancing base. Second, each scenario applies a city-specific market refinancing rate that is bounded and recorded in the output. Third, the resulting gap is passed through the common hierarchy of budget-buffer capacity, limited upper-level support, flexible-expenditure adjustment, and residual restructuring need. Fourth, revenue and expenditure outcomes are reported as scenario comparisons relative to the orderly-exit benchmark. This sequence is designed to avoid treating the final residual as a default event or treating the entire LGFV balance sheet as an immediate budget obligation. Several measurement boundaries deserve emphasis. The land-dependence and transfer-dependence indicators are comparative fiscal ratios generated from the harmonized panel. They are not legal classifications of earmarked revenue. The residual fiscal-capacity measure is similarly a model quantity: it is the difference between a narrow fiscal-resource measure and the protected floor, truncated at zero. It does not establish cash availability, debt-servicing priority, or a city government’s authority to reallocate funds across budget accounts. The paper therefore uses these indicators to compare the stress-test mechanics across cities, not to make an administrative assessment of any local government’s fiscal compliance or compensation policy. The city-year panel is intentionally retained in two forms in the replication materials: an observed-only version and a model-ready version. The observed-only version preserves raw missingness. The model-ready version uses transparent compatibility rules and limited imputations for panel construction, with source and quality flags. The national charts and city-profile calculations use the model-ready version, while the data audit files document coverage, missing values, unmatched issuers, and bond-deduplication decisions. This design allows users to reproduce the paper’s scenarios, alter the refinancing assumptions, or restrict analysis to observed-only records without changing the conceptual accounting framework.

B. Idealized Assumptions, Institutional Frictions, Public-Sector Compensation

The stress test is intentionally transparent, but this transparency requires a set of simplifying assumptions that are more idealized than the institutional environment in which local fiscal adjustment actually occurs. The framework treats the refinancing gap as an accounting pressure that can be allocated sequentially across modeled budget buffers, limited upper-level support, flexible expenditure adjustment, and residual restructuring need. This structure is useful because it makes the adjustment hierarchy explicit. However, it implicitly assumes that a nontrivial share of measured fiscal resources can be mobilized within the scenario horizon once core public services and debt-interest expenditure are protected. In practice, this assumption may be restrictive. Fiscal resources may be earmarked across budget accounts, tied to specific projects or transfer programs, subject to administrative approval, temporarily unavailable because of cash-flow timing, or politically difficult to reallocate. A city may therefore appear to possess positive modeled buffer capacity while facing a much narrower set of immediately deployable resources. Several other assumptions should be interpreted in the same spirit. First, the model uses publicly observed LGFV bond maturities as the refinancing base. This improves measurement transparency, but it does not capture the full timing and contractual structure of bank loans, trust products, supplier arrears, guarantees, leasing obligations, or other contingent liabilities. Second, the scenario design applies refinancing, land-revenue, transfer, and expenditure assumptions through common rules. Actual adjustment may instead be highly path dependent: a city with a weak land market may experience a sharper decline in land-transfer revenue than implied by the scenario, while another city may obtain temporary liquidity support through provincial coordination, asset transfers, policy-bank lending, debt swaps, or project restructuring. Third, the framework does not model feedback loops explicitly. A deterioration in LGFV refinancing conditions may weaken local investment, construction activity, land demand, and tax bases; these effects could in turn further reduce fiscal capacity and make subsequent refinancing more difficult. Conversely, credible restructuring arrangements or targeted support could improve market confidence and partially offset the initial shock.

These limitations define a productive agenda for future work rather than invalidate the comparative logic of the present exercise. A more realistic framework could combine monthly or quarterly debt-maturity schedules with cash-balance information, account-level budget execution data, earmarking restrictions, and detailed information on the legal priority of different expenditure categories. It could also model local governments, LGFVs, banks, land markets, and upper-level fiscal authorities as jointly interacting balance sheets rather than as separate adjustment channels. Such a framework would allow refinancing conditions, land prices, transfers, and expenditure commitments to evolve endogenously over

time. In particular, future research could distinguish between a city that has sufficient aggregate fiscal resources but faces a short-run liquidity mismatch and a city whose long-run fiscal capacity is genuinely insufficient even after restructuring and support are considered. The same distinction is important when interpreting the possible consequences of financing stress for workers within the public sector. The residual fiscal-capacity measure in this paper is not a wage, payroll, benefit, or personnel-compensation variable. It cannot establish whether civil servants, public-institution employees, local public-service workers, or LGFV employees will experience salary reductions, delayed payments, changes in bonuses, or changes in employment conditions. Compensation is governed by institutional rules that are not observed in the city-level fiscal panel, including staffing quotas, payroll classifications, central and provincial wage policies, earmarked transfers, departmental budget arrangements, and the distinction between general-government units, public institutions, and state-owned or LGFV-affiliated enterprises.

Nevertheless, government-financing stress may affect public-sector compensation through several plausible fiscal channels. Basic wages and statutory employer contributions are likely to be more rigid than discretionary spending in the short run, especially where local governments seek to preserve essential public services and administrative continuity. Adjustment may therefore occur first through margins that are less legally or politically protected, such as delayed recruitment, staff attrition, reduced temporary hiring, lower performance-related compensation, smaller local allowances, deferred reimbursements, reduced welfare supplements, or a reallocation of personnel toward core service functions. These channels need not imply an immediate reduction in nominal base pay. Rather, they suggest that the incidence of fiscal stress may appear through slower compensation growth, weaker bonus pools, fewer promotion or hiring opportunities, greater work intensity, or uneven treatment across departments and localities. The exposure may also differ sharply across groups commonly described as “within the system.” Civil servants whose compensation is financed through the general budget may be relatively insulated where payroll obligations are prioritized. Employees in public institutions, including schools, hospitals, and other service providers, may face different pressures because their funding can combine general-budget appropriations, earmarked transfers, service revenue, and project-specific funds. Employees of LGFVs and related local state-owned entities may be more directly exposed to refinancing conditions because their compensation and employment stability depend more closely on enterprise cash flow, asset operations, project payments, and access to market funding. Treating these groups as a single homogeneous category would therefore obscure potentially important differences in the transmission of local financing stress. A more complete empirical analysis of compensation effects would require administrative payroll records, staffing and recruitment data, information on performance pay and allowances, social-insurance contribution records, and balance-sheet data for local public institutions and LGFVs. Linking such information to city-level maturity schedules and refinancing shocks could help distinguish whether fiscal stress primarily affects wage levels, wage growth, employment margins, benefit provision, or payment timing. The present paper does not make such claims. Its narrower contribution is to identify where a transition away from implicit LGFV guarantees may compress modeled fiscal room and thereby create conditions under which personnel-related adjustment becomes more likely to be considered alongside project reprioritization, restructuring, and targeted fiscal support.